RICHFIELD GRADUATE INSTITUTE OF TECHNOLOGY

IT PROJECT 700 PHASE 3

EMERGEX: a campus-linked student accommodation search and viewing-request system for Richfield students.

Group members

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3.1. Introduction

This phase builds on the need, feasibility and requirements established in Phase 2. Its purpose is to examine how Richfield students currently find off-campus accommodation, identify the weaknesses in that process, and define in detail what the proposed EMERGEX system must do and which data rules it must enforce.

Richfield operates eight campuses across four provinces (Gauteng, KwaZulu-Natal, Western Cape and Limpopo) but does not own student residences. Students are referred informally to off-campus housing, with no central way to compare prices, check distance to campus or confirm whether a property has been inspected. EMERGEX addresses this as a campus-linked accommodation search and viewing-request system.

During Phase 2, the scope was refined from the Phase 1 proposal. Online booking, Instant Book and payment-gateway processing were moved to future scope ("Won't (Yet)") so the team could deliver a complete, well-tested core within the semester. NSFAS-related features were removed after the preliminary investigation confirmed that NSFAS funds only public universities and TVET colleges, so Richfield students are not eligible. This analysis follows the confirmed Phase 2 scope: campus and listing search, listing details, a map view, a shortlist, a move-in cost estimate and a validated viewing-request form (FR1–FR10, NFR1–NFR6).

The sections that follow cover the information-gathering methods used (3.2), the existing system (3.3), data integrity and constraints (3.4), weaknesses of the current approach (3.5), the functional and non-functional requirements of the proposed system (3.6 - 3.7) and the refined data models (3.8).

3.2. Information-Gathering Methodology

We combined four methods (triangulation) so that findings from one method could be checked against another. Secondary research established the wider context, structured observation examined the channels students use today, participatory sessions turned the findings into requirements, and semi-structured interviews were planned to validate them with Richfield students.

Method	Source	Main finding
Document analysis	News reports, police warnings, academic studies, Richfield website	Severe bed shortage; widespread social-media scams; students prioritise safety, cost and distance
Structured observation	Richfield website, Gumtree, Facebook groups, WhatsApp referrals	Fragmented search; no campus filter; no verification status
Participatory sessions	Six team members in stakeholder roles	MoSCoW scope and FR1–FR10
Interviews	Richfield students (planned)	To validate the findings above

3.2.1. Document Analysis (Secondary Research):

- Shortage of Beds: South Africa faces a shortfall of more than 500 000 student beds, so many students must find private housing themselves (Johnson, 2025). This applies even more strongly to Richfield, which operates no residences.

- Scams and unverified listings: Facebook and WhatsApp have become common channels for fake listings that promise affordable rooms near campus and pressure students into urgent deposits (Johnson, 2025). Universities and police report scammers advertising on Facebook Marketplace and disappearing once the deposit is paid (Sowetan, 2025). Police have warned against paying deposits into private accounts advertised online, noting that some scammers use professional-looking fake websites (SABC News, 2017). Students are advised to verify listings through their institution's housing office before paying (Majadibodu, 2026); Richfield has no such office.

- What students value: A Cape Town study found that students prioritise convenience, safety, cost and privacy, with monthly rent and then distance from campus among the most influential factors (Edwards, 2019). A University of Johannesburg study found that students look for secure residences within walking distance of campus with reliable internet (Kobue, Oke and Aigbavboa, 2017).

- Richfield's context: Richfield describes its eight campuses as close to public transport, shops and student accommodation (Richfield, n.d). Its Pretoria campus page notes that students commute from areas such as Hatfield, Brooklyn, Muckleneuk, Sunnyside and the CBD (Richfield, n.d), so accommodation is found by students in surrounding suburbs, not provided by the institution.

3.2.2. Structured Observation

We observed the public channels a Richfield student would realistically use, repeating one task on each: find an affordable room near a specific Richfield campus and arrange to view it.

Channel	What was observed	Implication for EMERGEX
Richfield website	Campus pages describe locations and surroundings but list no accommodation, landlords or inspected properties	Students leave the institution's site to search elsewhere
Gumtree	Adverts mention nearby institutions in free text (e.g. a Durban advert naming Richfield College) but cannot be filtered by campus or distance	No reliable campus-based search or distance information

	(Gumtree South Africa, n.d.)	
Facebook Marketplace and groups	Main informal channel; named in news reports as the main source of fake listings (Johnson, 2025; Sowetan, 2025; Majadibodu, 2026)	No verification status; high scam risk
WhatsApp referrals	Word of mouth between students; no record of price, terms or inspection	No comparison and no audit trail

On every channel, the student had to switch platforms to compare prices, estimate distance, and contact the landlord, and no channel showed whether a property had been inspected.

3.2.3. Participatory Sessions

Team members took on the viewpoints of the system's stakeholders (a student searching for a room, a parent or guardian paying for it, and a staff member maintaining listings) and walked through the search-to-viewing process together. These sessions produced the MoSCoW prioritisation and FR1–FR10 submitted in Phase 2 and identified the need for a move-in cost estimate (FR8) so that parents see the full deposit and fee costs before a decision is made.

3.2.4. Interviews

Semi-structured interviews with Richfield students were planned to confirm these findings. At the time of submission, the interview responses were still being consolidated, so the findings in this report rest on document analysis, observation and participatory sessions. The interview guide is:

1. How did you find your current accommodation?
2. Which websites, apps or groups did you use, and which was most useful?
3. What information did you need before deciding to view a room?
4. How far is your accommodation from campus, and how do you travel?
5. Have you ever seen a listing you thought was fake, or been asked for a deposit before viewing?
6. Did your parent or guardian see the costs and lease terms before you committed?
7. Would a list of inspected properties linked to your campus change how you search?

3.3. Analysis of the Existing System

Richfield does not operate an accommodation system. The existing system is an informal process that students piece together from unconnected channels: referrals from campus staff and other students, classified sites such as Gumtree, Facebook Marketplace and student-housing groups, and direct WhatsApp or telephone contact with landlords. Each channel works on its own, and no single party holds a complete record of the search, the property or the decision.

3.3.1. Stakeholders in the current process

Stakeholder	Role in the current process
Student	Searches for a room, compares options, contacts landlords and arranges viewings, often while new to the city
Parent or guardian	Usually pays the deposit and rent, but is typically involved only after the student has chosen a room
Landlord or agent	Advertises rooms on one or more channels and responds to enquiries individually
Richfield campus staff	Answer general queries and refer students to off-campus housing informally; they do not list, inspect or endorse properties
Fellow students	Share recommendations and contacts by word of mouth and in WhatsApp groups

3.3.2. Current Process (as-is)

No.	Activity	Who	Channel Used	Information available at this step
1	Realises accommodation is needed (new registration or new year)	Student	–	Campus location only
2	Asks campus staff or other students for suggestions	Student, staff, peers	In person, WhatsApp	Informal names and contact numbers
3	Searches online for available rooms	Student	Gumtree, Facebook, rental sites	Advert text and photos; price sometimes shown
4	Compares options	Student	Manually, with a separate map app for distance	Whatever each advert happens to include
5	Contacts landlords and requests a viewing	Student, landlord	WhatsApp, phone call	Replies in personal chats
6	Views the room, if a viewing is possible	Student, landlord	In person	Physical condition of the property
7	Shares the choice with parent or guardian; deposit is paid	Student, parent	Phone call, bank transfer	Deposit and lease terms, often only now
8	Moves in	Student, landlord	In person	Lease, if one is issued

Figure 1 below shows the same process as a swim lane activity diagram. The parent or guardian lane stays empty until the room has already been chosen.

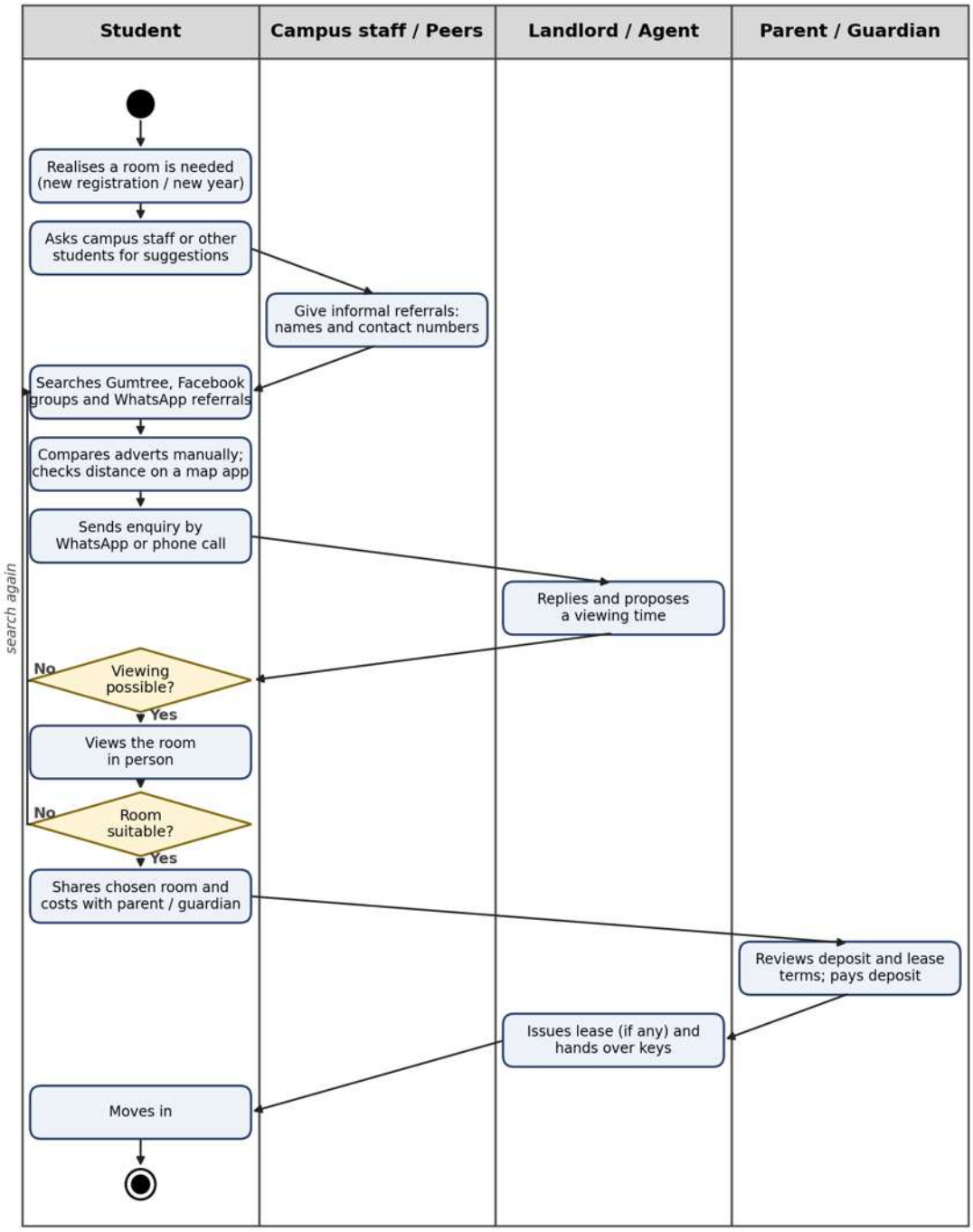


Figure 1: The As-is activity diagram of how a Richfield student currently finds off-campus accommodation

3.3.3. Information handled in the current process

- Property details (rent, room type, amenities): are typed freely into adverts, so the same detail is described differently, or left out, from one listing to the next.

- Distance to campus is rarely stated; students check it themselves in a separate map application.
- Inspection or verification status is not recorded anywhere; students are advised to verify landlords themselves before paying (SABC News, 2017; Majadibodu, 2026).
- Enquiries and viewing arrangements sit in personal WhatsApp chats and call logs, with no reference number, status or history.
- Total move-in cost (deposit, admin fee and rent) is usually only fully known once the student is close to committing.

As a result, nobody in the process (student, parent, landlord or campus) sees a complete, consistent picture of the options and their cost. The weaknesses this causes are analysed in Section 3.5.

3.4. Data Analysis (Data Integrity and Constraints)

EMERGEX holds nine tables, each with keys, types and rules that keep its data accurate; the analysis also found six rules the prototype does not yet fully enforce (3.4.5).

3.4.1. Classification of Data

Type	Tables	Maintained by	Characteristics
Reference data	campuses, room_types, amenities, listings, listing_amenities, faqs	Staff or administrator	Changes rarely; read on almost every page
Captured data	viewing_requests, viewing_request_rooms	Students, through the viewing-request form	Grows over time; holds personal information protected under POPIA
Session data	shortlists	The student's browser (FR5)	Temporary; stored server-side only once student logins exist (Could-have)

3.4.2. The Types of Integrity Enforced

Integrity type	Meaning	How EMERGEX enforces it
Entity integrity	Every row is uniquely identifiable	A primary key on every table; composite keys on link tables
Referential integrity	A foreign key always points to an existing row	Foreign-key constraints; listings are deactivated, not deleted
Domain integrity	Every value has the right type and range	Data types, NOT NULL and server-side validation (FR7)
Business-rule integrity	Data obeys the rules of the process	Rules BR-01 to BR-14, checked in PHP before any write

3.4.3. Constraints Per Table

Table	Primary key	Key data types and constraints

campuses	id VARCHAR(40)	name, city, province VARCHAR(120) NOT NULL; lat, lng DECIMAL(10,7) NOT NULL; confirmed TINYINT(1) default 0; beds INT UNSIGNED default 0
room_types	id VARCHAR(40)	label VARCHAR(120) NOT NULL
amenities	id VARCHAR(40)	short VARCHAR(80) and label VARCHAR(160) NOT NULL
listings	slug VARCHAR (120)	campus_id → campuses and room_type → room_types (ON UPDATE CASCADE); area at suburb level only; lat, lng NOT NULL; rent INT UNSIGNED in whole rand; verified default 0; is_active default 1 (soft delete); created_at and updated_at set by the server; indexed on campus, rent and active status
listing_amenities	(listing_slug, amenity_id)	Both foreign keys, ON DELETE CASCADE; composite key prevents duplicate links
faqs	id INT AUTO_INCREMENT	question VARCHAR(300) and answer TEXT NOT NULL; is_active default 1
viewing_requests	id INT AUTO_INCREMENT	reference VARCHAR(20) UNIQUE; full_name, email, phone, campus_id, preferred_date (DATE), move_in_month (CHAR(7), YYYY-MM) NOT NULL; guardian_email and notes (≤ 500 characters) optional; campus_id → campuses; status default "new"; created_at default current time; indexed on campus, status and date
viewing_request_rooms	(request_id, listing_slug)	Both foreign keys; deleting a request deletes its room rows
shortlists	id INT AUTO_INCREMENT	listing_slug → listings; UNIQUE (owner_key, listing_slug); used only once logins exist

3.4.4. Business Rules

Rule	Description	Enforcement	Requirement
BR-01	Only active listings appear in search results	is_active = 1 in every listing query	FR2

BR-02	A listing's inspected status is shown exactly as stored, never assumed	Badge drawn from the verified field only	FR3, NFR5
BR-03	Campuses whose location is unconfirmed are labelled as such	"Location to confirm" badge when confirmed = 0	FR1, NFR5
BR-04	Every listing belongs to exactly one campus and one room type	NOT NULL foreign keys	FR2, FR3
BR-05	A student may shortlist at most four listings	Limit checked before adding; duplicates ignored	FR5
BR-06	A viewing request must name at least one listing	Required; not yet enforced server-side (3.4.5)	FR6
BR-07	Rooms in a request must be active and belong to the selected campus	Partly enforced: rooms must exist, but are not matched to the campus (3.4.5)	FR6, FR7
BR-08	Phone must be a South African mobile number (0 or +27, then 6, 7 or 8, then 8 digits)	Server-side pattern check	FR7
BR-09	Preferred viewing date cannot be in the past	Server-side date check	FR7
BR-10	Each request gets a unique server-generated reference (RF-YYMMDD-XXXX)	Generated in PHP; UNIQUE constraint	FR6
BR-11	A request and its rooms are saved together or not at all	One database transaction	FR6
BR-12	Only data needed to arrange a viewing is collected	No ID numbers, banking or funding documents	NFR5, POPIA
BR-13	Move-in cost = rent for the lease term (less 5% upfront or 2% termly, on rent only) + deposit (rent × chosen deposit months) + fixed R 950 admin fee, in whole rand	Server-side calculation with fixed system constants	FR8
BR-14	A listing's location is shown at suburb level, never as a street address	area holds the suburb only	NFR5, POPIA

3.4.5. Findings from the Data Analysis

a) Model Documentation Gaps: The Phase 2 ERD omitted attributes and a table that the requirements depend on: listing coordinates (FR3, FR4); move_in_month, guardian_email and notes (FR6); and the faqs table (FR10). All of these already exist

in the prototype's schema and are added to the refined model in Section 3.8. FR8 needs no new fields, because the deposit and admin fee are fixed system settings (BR-13).

b) Integrity Rules not yet fully enforced: Comparing the business rules with the prototype found the following, carried into Phase 4 as design corrections:

No.	Finding	Risk	Recommended correction
1	A viewing request can be saved with no rooms selected	Requests the campus cannot act on	Reject the form if no listing is selected (BR-06)
2	Selected rooms are not checked against the selected campus	A request for one campus naming another campus's room	Server-side campus match (BR-07)
3	move_in_month and campus are accepted without checking them against the allowed options	Invalid values stored, especially when no database constraint applies	Validate against the generated month list and the campus list
4	status is free text with no database restriction	Inconsistent statuses once an admin view is added	Restrict to new, contacted, viewed, closed (ENUM or CHECK)
5	Deleting a listing removes it from past viewing requests (ON DELETE CASCADE)	Loss of request history	Use soft delete only and change the rule to ON DELETE RESTRICT
6	Full-name length is not checked against the 160-character column	Truncation or a database error	Add a server-side maximum length

3.5. Weaknesses of the Current System

The current process has eight weaknesses, all traceable to one root cause: there is no single, campus-linked source of structured and verified accommodation information. Each weakness below is linked to the as-is step where it occurs (Table 3.3.2) and to the EMERGEX requirement that addresses it.

No.	Weakness	Root cause	Where it occurs (as-is step)	Evidence	EMERGEX response
W1	Search is fragmented across many channels	No central platform for Richfield students	Steps 2–4	Observation 3.2.2	One site for every campus and listing (FR1, FR2)
W2	No search by campus and no distance information	Adverts are not linked to a campus or a location	Step 4	Observation 3.2.2; students rank distance highly	Campus filter, walking and driving times,

				(Edwards, 2019; Kobue, Oke and Aigbavboa, 2017)	map (FR2, FR3, FR4)
W3	No way to tell a genuine or inspected property from a fake one	Inspection status is never recorded	Steps 3–6	Scam reports and police warnings (Johnson, 2025; Sowetan, 2025; SABC News, 2017; Majadibodu, 2026)	Inspected badge shown only from stored data (FR3, BR-02)
W4	Property information is inconsistent and incomplete	Free-text adverts with no required fields	Steps 3–4	Observation 3.2.2	Structured rent, room type and amenities for every listing (FR3, BR-04)
W5	Full move-in cost is known only late, and parents are involved last	No tool to estimate costs; guardians are outside the process	Step 7	As-is diagram (Figure 1); Phase 2 identification of need	Move-in cost estimate and optional guardian email (FR8, FR6)
W6	Enquiries are untracked, with no reference or status	Contact happens in personal chats	Step 5	Observation 3.2.2	Viewing request with a unique reference and status (FR6, BR-10)
W7	Students are pressured to pay a deposit before viewing	Scarcity, and no viewing-first process	Steps 5–7	Scam reports (Johnson, 2025; Sowetan, 2025; Majadibodu, 2026)	A free viewing request that takes no payment; payment stays out of scope
W8	Personal details are shared over uncontrolled channels	No secure, validated form	Step 5	POPIA requirements (South African Government, 2013)	Validated form protected against cross-site request forgery; minimal data (FR7, FR9, BR-12)

W3 and W7 carry the highest risk, because together they expose students to losing a deposit on a property that does not exist. W5 is the weakness most specific to Richfield, because the parent or guardian who pays is the last person to see the costs.

3.6. Analysis of the Proposed System (Functional Requirements)

EMERGEX replaces steps 2 to 5 of the current process with one website. A student picks a campus, compares structured listings with distance and inspection status, estimates the full move-in cost, and sends one tracked viewing request that can name a parent or guardian.

3.6.1. Actors

Actor	Type	Interaction with EMERGEX
Student	Primary	Searches, views listings, shortlists, estimates costs and submits viewing requests
Parent or guardian	Secondary	Optional email recorded with the request so staff can include them when following up
Staff (administrator)	Supporting	Maintains campuses, listings, amenities and FAQs; follows up viewing requests. No admin screen yet: data is maintained through the database and seed script (Could-have)

The prototype stores the guardian email but sends no emails yet. The site's FAQ says guardians "receive the same confirmation"; under NFR5 (Honesty) that answer must be reworded until email sending is built.

3.6.2. Functional Requirements in Detail

Priorities follow the Phase 2 MoSCoW analysis.

ID	Requirement	Input	Processing	Output	Priority
FR1	Display every campus with city, province, confirmed status and bed count	None	Read all campuses in sort order	Campus cards; "Location to confirm" badge where unconfirmed	Must
FR2	Search listings by campus, maximum rent, room type and amenities	Filter selections	Query active listings matching all filters (BR-01)	Matching listings, or an empty-state message	Must
FR3	Show rent, room type, inspected status,	Listing selected	Read listing, room type and amenities; calculate	Listing page with badges and travel times	Must

	amenities and walking and driving distance for each listing		distance from listing to campus coordinates		
FR4	Plot listings and their campus on an interactive map	Campus or search result	Pass coordinates to the map library (Leaflet with OpenStreetMap)	Map with campus and listing pins	Must
FR5	Shortlist up to four listings, kept across page reloads	Add or remove action	Check the four-item limit; store in the browser	Shortlist drawer; listings carried into the viewing request	Should
FR6	Accept a viewing request with contact details, preferred date, move-in month and selected listings	Form fields	Validate (FR7), generate reference, save request and rooms in one transaction (BR-10, BR-11)	Confirmation showing the reference	Must
FR7	Validate all viewing-request input server-side, including South African mobile format	Submitted form	Check each field against BR-06 to BR-09	Field-level error messages, inputs kept	Must
FR8	Estimate total move-in cost for a chosen lease length and payment plan	Rent, lease months, deposit months, plan	Apply BR-13	Amount due now, instalment and total, in rand	Should
FR9	Protect the viewing-request form against cross-site request forgery	Hidden session token	Compare the submitted token with the session token	Request rejected if the tokens differ	Must
FR10	Keep showing campus, listing and FAQ content with no	None	Fall back to built-in sample data	Same pages, with a sample-data notice	Must

	database connected				

3.6.3. Use Cases

The use-case diagram (Figure 2) shows six use cases: UC1 Search listings, UC2 View listing details, UC3 Manage shortlist, UC4 Estimate move-in cost, UC5 Submit viewing request and UC6 Maintain reference data. UC5 is the main capture point of the system and is described in full below.

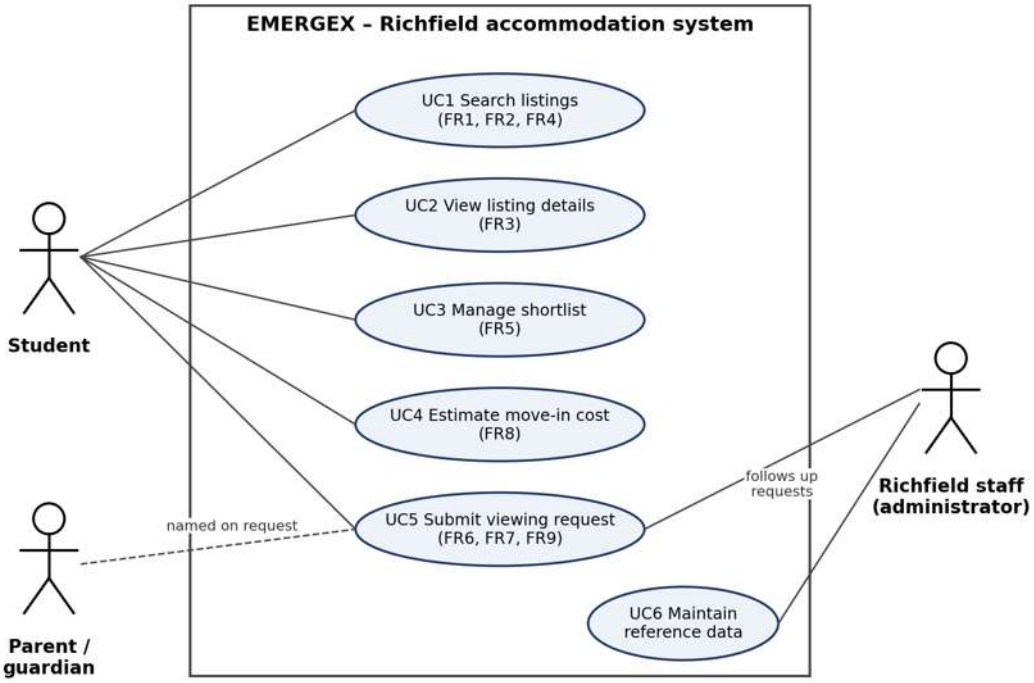


Figure 2: The Use-case diagram for the proposed EMERGEX system.

Field	UC5 Submit viewing request
Actor	Student (primary); parent or guardian (named on the request)
Requirements	FR6, FR7, FR9
Pre-condition	The student has found at least one listing from search, a listing page or the shortlist.
Main flow	1. Student opens the viewing-request form; listings from the shortlist or listing page are pre-selected. 2. Student enters full name, email, mobile number, optional guardian email, campus, preferred date, move-in month and optional notes, and confirms the selected rooms. 3. Student submits. 4. System checks the form token (FR9). 5. System validates every field (FR7).

	6. System generates a unique reference (BR-10). 7. System saves the request and its rooms in one transaction (BR-11). 8. System shows a confirmation with the reference.
Alternative: invalid input	At step 5, the form is shown again with a message beside each invalid field, and the student's entries are kept
Alternative: expired session	At step 4, the request is rejected with "Your session expired. Please try again."
Alternative: database unavailable	At step 7, the system still validates and shows a reference, but clearly states the request was not saved (FR10)
Post-condition	One viewing request with status "new" and at least one linked room exists in the database

3.7. Non-Functional Requirements

Each Phase 2 quality requirement now has a measurable acceptance criterion, and the analysis adds two requirements (NFR7, NFR8) that came out of the data analysis in 3.4. The last column records what the team's prototype test runs reported.

ID	Category	Requirement	Measurable criterion	How verified	Prototype status
NFR1	Performance	Pages render server-side with no client build step, on ordinary shared hosting	Runs with only PHP 8.0 or newer; no package install or build step; every page returns HTTP 200	Fresh install on a clean machine; route checks	Met: all routes return 200
NFR2	Security	Every database query uses a parameterised statement; input is never joined into SQL	0 queries built by string concatenation; injection attempts on every URL parameter leave the tables intact	Code review of the single query file; injection tests	Met
NFR3	Portability	Runs unmodified on MySQL/MariaDB or SQLite through PDO	Same code passes the same checks on both drivers; only the configuration file changes	Run the acceptance checks on each driver	Met: both schemas provided

NFR4	Accessibility	WCAG 2.2 AA colour contrast; every control keyboard-operable	Text contrast at least 4.5:1; every form, filter and drawer usable with keyboard only	Contrast checker; keyboard-only walkthrough	To be verified in Phase 5
NFR5	Honesty	Never present sample data as real, or claim NSFAS eligibility or an affiliation not held	Sample-data notice and "not affiliated" footer on every page; 0 NSFAS claims	Page-by-page content review	Partly met: FAQ wording on guardian emails to correct (3.6.1)
NFR6	Compatibility	No horizontal overflow from 360 px to 1920 px wide	0 px horizontal scroll at 360, 390, 414, 768, 1024, 1280 and 1920 px	Browser test at each width	Met
NFR7	Privacy (new)	Personal data is limited to what a viewing needs and protected under POPIA	Only the fields in BR-12 collected; credentials and data dumps never committed to the repository; a retention period agreed before go-live	Schema review; repository check	Partly met: retention period still to agree
NFR8	Security (new)	Every value shown on a page is escaped, and errors are logged rather than shown	A script tag typed into any field is displayed as text; a database failure shows a friendly message, never a stack trace	Test with script input; test with the database stopped	Met

3.8. Data Modelling for the Proposed System

The refined model completes the nine tables named in Phase 2 (its diagram drew eight) by adding the faqs table and the attributes the requirements need (3.4.5a); the context and Level 1 data-flow diagrams show how data moves between students, staff and these tables.

3.8.1. Changes from the Phase 2 data model

Table	Change	Reason
listings	Added lat, lng, blurb, image_url, created_at, updated_at	Map and distance (FR3, FR4); listing page content; audit of changes
viewing_requests	Added guardian_email, move_in_month, notes	FR6 and the guardian visibility need (W5)
faqs	Added as the ninth table	FR10 FAQ content
campuses, room_types, amenities	Added sort_order	Fixed display order on every page
shortlists	Added created_at	Ordering saved rooms once logins exist

3.8.2. Refined entity-relationship diagram

Figure 3 shows the refined ERD. A campus has many listings and many viewing requests; a listing has one room type and many amenities through listing_amenities; a viewing request names one or more listings through viewing_request_rooms. faqs stands alone.

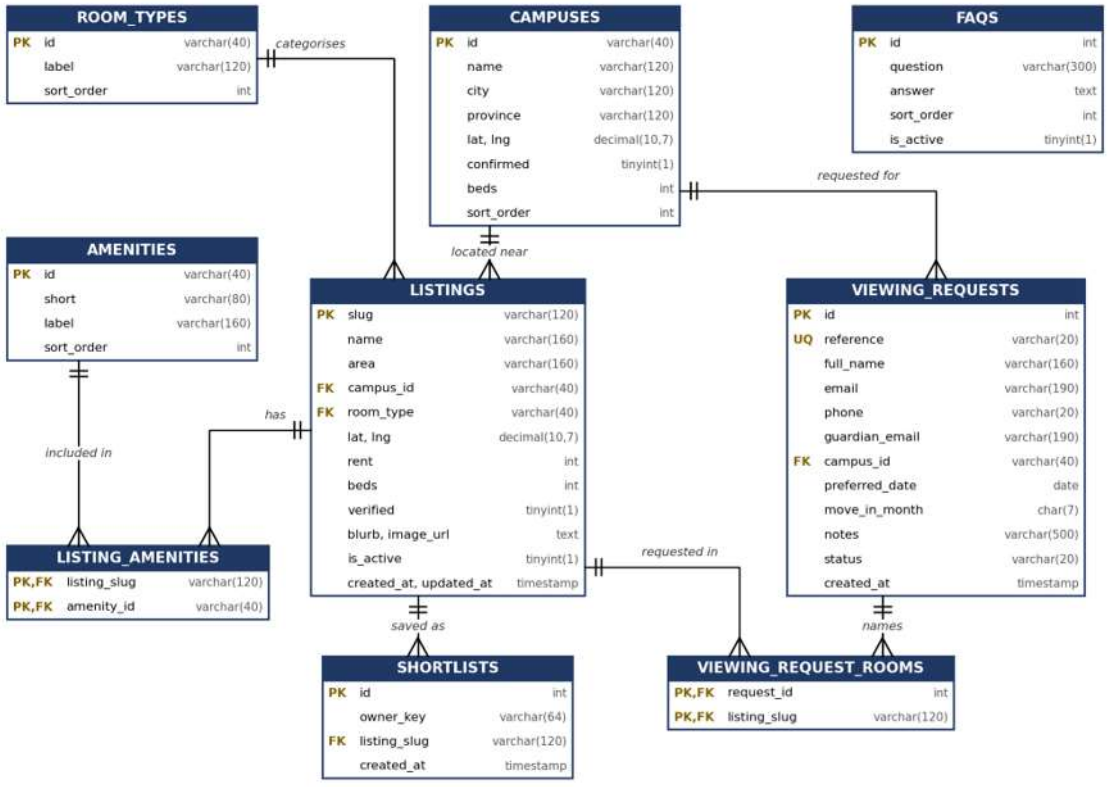


Figure 3: The Refined entity-relationship diagram (crow's foot notation)

3.8.3. Data dictionary (captured data)

Reference tables are defined in 3.4.3. The table that holds personal information is defined in full here.

Field	Type	Null	Key	Description and rule
id	INT UNSIGNED, auto-increment	No	PK	Internal identifier
reference	VARCHAR(20)	No	UNIQUE	RF-YYMMDD-XXXX, shown to the student (BR-10)
full_name	VARCHAR(160)	No		At least 2 characters
email	VARCHAR(190)	No		Valid email address
phone	VARCHAR(20)	No		South African mobile number (BR-08)
guardian_email	VARCHAR(190)	Yes		Valid email address when given
campus_id	VARCHAR(40)	No	FK → campuses	Campus the student attends
preferred_date	DATE	No		Not in the past (BR-09)
move_in_month	CHAR(7)	No		YYYY-MM, one of the next 12 months
notes	VARCHAR(500)	Yes		Optional, at most 500 characters
status	VARCHAR(20)	No		new, contacted, viewed or closed; default new
created_at	TIMESTAMP	No		Set by the server at insert

viewing_request_rooms links each request to its listings with a composite primary key (request_id, listing_slug).

3.8.4. Data-flow diagrams

The context diagram (Figure 4) shows EMERGEX as one process exchanging data with two external entities: the student and Richfield staff. The parent or guardian appears only as data (the optional guardian email), because the system does not yet send them anything. The Level 1 diagram (Figure 5) breaks the process into six sub-processes and four data stores; the shortlist is kept in the student's browser, so it has no data store.

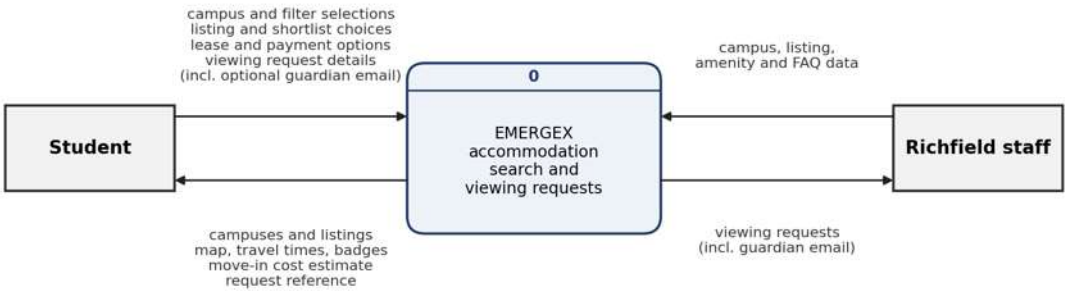


Figure 4: The Context-level data-flow diagram

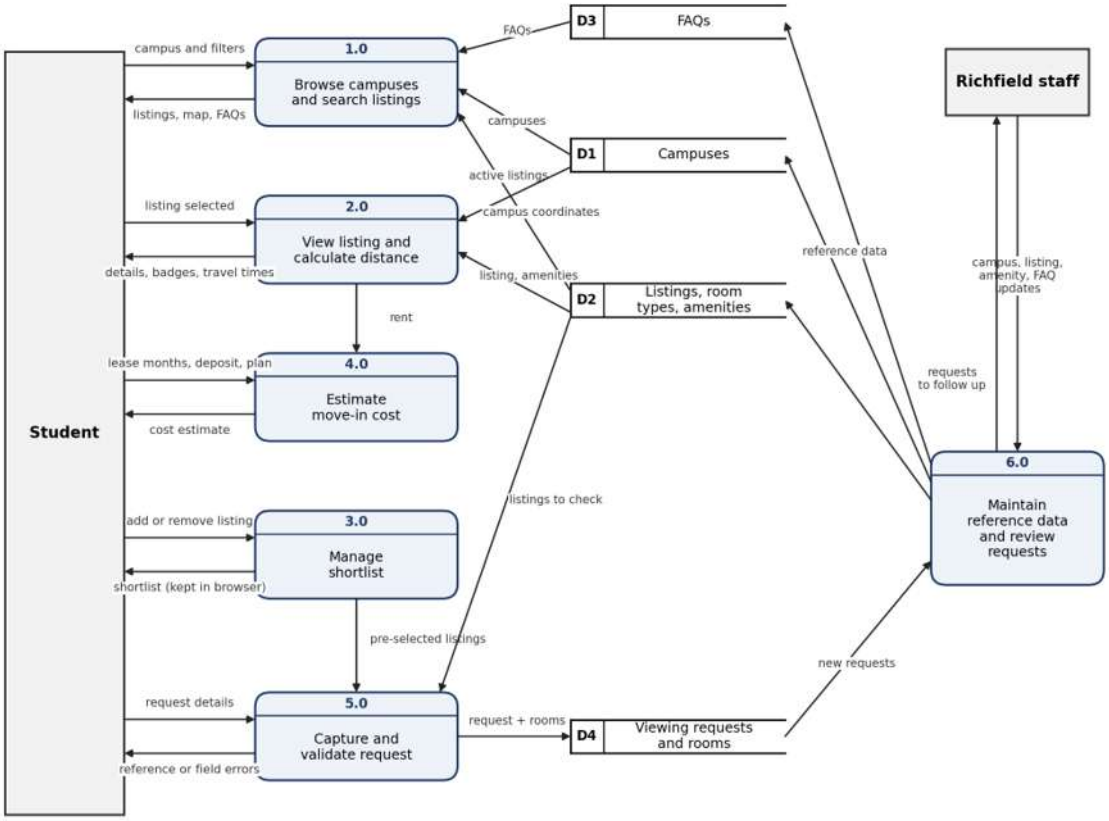


Figure 5: The Level 1 data-flow diagram

References

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